The area enclosed by a curve in the complex plane defined by a parametric representation $z(\theta) = x(\theta) + i\, y(\theta)$, where $x(\theta)$ and $y(\theta)$ are the real and imaginary parts of the curve, respectively, is given by the integral

$$A = \frac{1}{2} \int_0^{2\pi} x(\theta)\, \dot{y}(\theta) - y(\theta)\, \dot{x}(\theta) d\,\theta \tag{1}$$

For the polynomials $b_n(z)$ evaluated on the unit circle $z = e^{i\theta}$, the curve is parametrized by

$$z(\theta) = b_n(e^{i\theta}) = x(\theta) + i\, y(\theta) \tag{2}$$

with θ ranging from 0 to $2\,\pi$. The derivatives of $x(\theta)$ and $y(\theta)$ with respect to θ are needed for the integral. These are given by

$$\dot{x}(\theta) = \frac{d}{d\,\theta} \Re\{b_n(e^{i\theta})\} \tag{3}$$

$$\dot{y}(\theta) = \frac{d}{d\,\theta} \Im\{b_n(e^{i\theta})\} \tag{4}$$

To calculate the area A without explicitly using the real and imaginary part operators, we can utilize the properties of complex conjugates. Considering $z(\theta) = b_n(e^{i\theta})$ and its complex conjugate derivative, the area can be expressed as

$$A = \frac{1}{2} \int_0^{2\pi} \Im(z(\theta)\overline{\dot{z}(\theta)})\, d \tag{5}$$

or, equivalently,

$$A = \frac{1}{4\,i} \int_0^{2\pi} z(\theta)\overline{\dot{z}(\theta)} - \overline{z(\theta)}\dot{z}(\theta)\, d\,\theta \tag{6}$$

This formula integrates the product of the function and the conjugate of its derivative, effectively capturing the interaction of the real and imaginary components without separating them explicitly.

The area calculation may require symbolic or numerical methods depending on the complexity of the functions involved.

1 Green's Theorem

Green's theorem relates the line integral of a vector field around a simple, closed curve C to a double integral over the plane region D bounded by C. It is stated as

$$\oint_C (P\,d\,x + Q\,d\,y) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) d\,A \tag{7}$$

where P and Q are functions of x and y defined on an open region containing D and have continuous partial derivatives there, dx and dy are the differentials of x and y, and dA is the differential area element.

Proof:

The proof of Green's theorem can be broken down into two parts: one for the case when $Q=0$ and P is a function of y only, and the other for when $P=0$ and Q is a function of x only. The general case follows from the linearity of the integral and partial derivatives.

Case 1: $Q=0$ *and* $P=P(y)$

Consider a region D in the plane bounded by the curve C. The line integral of Pdx around C is zero since P depends only on y, and over a closed path, the total change in x is zero. Thus,

$$\oint_C P dx = 0 \tag{8}$$

Case 2: $P=0$ *and* $Q=Q(x)$

For a region D bounded by C, the line integral of $Q\,dy$ can be expressed as the change in Q along y, which, by the Fundamental Theorem of Calculus, is equal to the integral of its derivative with respect to y over the region D. Thus,

$$\oint_C Q\,dy = \iint_D \frac{\partial Q}{\partial y}\,dA \tag{9}$$

The general case of Green's theorem follows by considering vector fields where both P and Q are non-zero and applying the superposition principle. By dividing the region D into smaller regions where either $P=0$ or $Q=0$, and then summing the contributions from each region, we obtain the formula stated at the beginning.